

Homework 2

2.3.4. (a.) The derivative of a distribution $f \in \mathcal{S}'$ is defined by the rule

$$\langle \partial f, \varphi \rangle = -\langle f, \partial \varphi \rangle, \quad \text{for } \varphi \in \mathcal{S}.$$

To compute the derivative of $\chi_{[a,b]}$, letting $\varphi \in \mathcal{S}$, we have

$$\langle \partial_x \chi_{[a,b]}, \varphi \rangle = -\langle \chi_{[a,b]}, \partial_x \varphi \rangle = -\int_a^b \partial_x \varphi = \varphi(a) - \varphi(b).$$

So the action of $\partial_x \chi_{[a,b]}$ is the same as $\delta_a - \delta_b$, so this is the distributional derivative.

(b.) Take $x_j = x$ without loss of generality. Again letting $\varphi \in \mathcal{S}(\mathbb{R}^2)$, we can write

$$\begin{aligned} \langle \partial_x \chi_{B(0,1)}, \varphi \rangle &= -\int_{B(0,1)} \partial_x \varphi(x, y) dV = -\int_{-1}^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \partial_x \varphi dx dy \\ &= \int_{\partial B(0,1)^-} \varphi dS - \int_{\partial B(0,1)^+} \varphi dS, \end{aligned}$$

where dS denotes a measure on the boundary $\partial B(0,1)$ and $-$ and $+$ denote left and right semicircles. This is the difference between the integral of φ over the left and right semicircles.

(c.) We compute the fourier transforms of $\sin(x)$ and $\cos(x)$, which are not integrable functions in the L^1 sense. We use the definition

$$\langle \mathcal{F}u, \varphi \rangle = \langle u, \mathcal{F}\varphi \rangle, \quad \text{for } \varphi \in \mathcal{S}.$$

We already know that $\widehat{(e^{2i\pi x \cdot a})} = \delta_a$ for $a \in \mathbb{R}$, so that $\widehat{(e^{ix})} = \delta_{\frac{1}{2\pi}}$. Moreover, $\sin(x)$ can be expressed as

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

so that by linearity of the fourier transform, we have

$$\mathcal{F}(\sin(x)) = \frac{1}{2i} \left(\delta_{\frac{1}{2\pi}} - \delta_{-\frac{1}{2\pi}} \right).$$

(d.) We need to compute

$$\langle \partial \log |x|, \varphi \rangle = -\langle \log |x|, \partial_x \varphi \rangle$$

for any $\varphi \in \mathcal{S}(\mathbb{R})$. Let $\varepsilon > 0$. Then

$$\int_{\varepsilon \leq |x|} \varphi(x) \frac{1}{x} dx = -\int_{\varepsilon \leq |x|} \partial_x \varphi(x) \log |x| dx$$

after integrating by parts, noting that the first integrand is integrable for any $\varepsilon > 0$. Moreover, the function $\log |x|$ is integrable on $\mathbb{R}$, so we can let epsilon go to zero in the final integral. So the limit converges for all φ :

$$u(\varphi) = \lim_{\varepsilon \rightarrow 0} \int_{\varepsilon \leq |x|} \varphi(x) \frac{1}{x} dx = -\int_{\mathbb{R}} \partial_x \varphi(x) \log |x| dx = \langle \log |x|, \partial_x \varphi \rangle =: \langle \partial_x \log |x|, \varphi \rangle.$$

We have proven the sequence of tempered distributions defining u converges to $\partial_x \log |x|$ in the weak* topology on $\mathcal{S}'(\mathbb{R})$, and so this is the derivative.

2.3.5. Let $\varphi \in C_0^\infty$ and $f \in \mathcal{S}$. Choose M so that $|\varphi| < M$ for all x .

(a.) Let

$$t_n = \frac{\tau^{-he_j} f - f}{h} - \partial_j f.$$

First, we can see that the quotient converges pointwise to $\partial_j f$. Fix $\varepsilon > 0$. To show uniform convergence, choose compact $K \subset \mathbb{R}^n$ such that $|f'(x)| < \varepsilon/2$ outside K . Then by the fundamental theorem of calculus,

$$|f(x + he_j) - f(x)| = \left| \int_0^h \partial_t f(x + te_i) dt \right| \leq \int_0^h |\partial_t f(x + te_i)| dt \leq h\varepsilon/2$$

for $x \in K^c$, given that t is small enough so $x + te_i$ also lies within K^c . Then for this same x , we have

$$\left| \frac{\tau^{-he_j} f - f}{h} - \partial_j f \right| \leq \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Moreover, within K , pointwise convergence implies uniform convergence, so we can choose h sufficiently small so that the quotient is as close to $\partial_j f$ as we want. Then we have proven uniform convergence after combining the control over both regions.

In fact, for any α , $\partial^\alpha(t_n)$ converges to zero uniformly by the exact same argument, since $\partial^\alpha f$ is schwartz as well.

Lastly, to show that $x^\alpha(t_n)$ converges to zero uniformly, note that we can choose M large so that $|x^\alpha f(x)| \leq (1 + |x|)^{-M}$ for all x . This gives the uniform bound outside a compact set K , and within K , x^α is bounded, so we still have pointwise and therefore uniform convergence.

(b.) Let $\varphi \in C_0^\infty$, so $|\varphi| \leq M$ for some $M > 0$. Then choose a compact set K so that $|f| < \frac{\varepsilon}{2M}$ for all $x \in K^c$. Within K , $\varphi_k(x)f(x)$ converges pointwise to f and thus uniformly since these functions are continuous. Outside of K , we have $|\varphi_k(x)f(x) - f(x)| \leq \frac{M\varepsilon}{2M} + \frac{\varepsilon}{2M} \leq \varepsilon$.

Now we show uniform convergence of the derivatives. We have

$$|\partial_j(\varphi_k f) - \partial_j f| \leq |\partial_j \varphi_k \cdot f| + |\varphi_k \partial_j f - \partial_j f|.$$

We will prove both of these terms are small individually. Since φ_k is flat near 1, for any α , $\partial^\alpha \varphi_k$ converges pointwise to 0 as $k \rightarrow \infty$. Then the first term can be made less than $\varepsilon/2$ if we choose k large enough so that within the support of φ_k , $|f| \leq \frac{\varepsilon}{2M}$. For the second term, the proof goes exactly the same as the previous step without the derivative.

For arbitrary partial derivatives, we could expand with the product rule, but every term containing a derivative of φ_k would vanish in a neighborhood of the origin and could be controlled as we did in the case of ∂_j . Then only the term with all derivatives hitting f survives, and

$$|\varphi_k \partial^\alpha f - \partial^\alpha f| \rightarrow 0$$

as in the previous case.

For any α , since $x^\alpha f$ is still a schwartz function, the proof is exactly the same here.

(c.) In the previous part we showed that $\varphi_k f \rightarrow f$ in the Schwartz space topology. Then it follows that $\widehat{\varphi_k f} \rightarrow \widehat{f}$ in $\mathcal{S}$ as well. In particular we have uniform convergence. Choose N_1 large enough that $\|\widehat{(\varphi_k f)} - \widehat{f}\|_{L^\infty} \leq \frac{\varepsilon}{4M}$ when $k > N_1$, and choose a compact set K so that $|\widehat{f}| \leq \frac{\varepsilon}{4M}$ for $x \in K^c$. Then by the triangle inequality, for all $k > N_1$, we have $|\widehat{(\varphi_k f)}| \leq \frac{\varepsilon}{2M}$ for $x \in K^c$. Thus

$$|\widehat{\varphi_k(\varphi_k f)} - \widehat{f}| \leq \varepsilon/2 + \frac{\varepsilon}{4M} \leq \varepsilon$$

Within the compact set K , we clearly have pointwise convergence since φ_k converges pointwise to 1 and $\widehat{(\varphi_k f)}$ converges pointwise to $\widehat{f}$. Then we can choose N_2 so that $|\varphi_k(\widehat{(\varphi_k f)}) - \widehat{f}| < \varepsilon$ on K when $k > N_2$, since K is compact.

Now set $N = \max\{N_1, N_2\}$. The difference will be less than epsilon for $k > N$ within and outside of K .

To show derivatives converge as well, write

$$\partial_j(\varphi_k(\widehat{(\varphi_k f)})) = \partial_j \varphi_k(\widehat{(\varphi_k f)}) + \varphi_k((-2\pi i x_j) \widehat{(\varphi_k f)}),$$

and note the similarity to part (b.). In general for ∂^α , we see again that the terms involving derivatives of φ_k will be zero around 1 and can be controlled like before. We can make them all small since we are only taking finite derivatives. This leaves only the term with all derivatives on f , but we can just replace f by $(-2\pi i x)^{-\alpha} f$ to return to the original case.

For convergence after multiplying by x^α for any α , choose a compact set K and N large so that $|x^\alpha(\widehat{(\varphi_k f)})| \leq \frac{\varepsilon}{2M}$ for some $C > 0$ for $x \in K^c$ when $k > N$ again using triangle inequality and the fact that $\widehat{(\varphi_k f)}$ converges to $\widehat{f}$ uniformly. Then we have

$$|\varphi_k x^\alpha(\widehat{(\varphi_k f)}) - x^\alpha \widehat{f}| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2M} \leq \varepsilon$$

for all $x \in K^c$ and all $k > N$.

We can get uniform bounds within K in exactly the same way as before. Then combine by taking the larger of the two N .

2.3.7. Let

$$g_N(\xi) = \int_{B(0,N)} f(x) e^{-2\pi i x \cdot \xi} dx.$$

Note that $g_N = (\chi_{B(0,N)} f)$. We show g_N converges to $\widehat{f}$ in $\mathcal{S}'$. This means showing $\langle g_N, \varphi \rangle \rightarrow \langle \widehat{f}, \varphi \rangle := \langle f, \widehat{\varphi} \rangle$ for any $\varphi \in \mathcal{S}$. We have

$$\begin{aligned} \langle g_N, \varphi \rangle &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \chi_{B(0,N)}(x) f(x) \varphi(\xi) e^{-2\pi i x \cdot \xi} dx d\xi \\ &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \chi_{B(0,N)}(x) f(x) \varphi(\xi) e^{-2\pi i x \cdot \xi} d\xi dx \\ &= \int_{\mathbb{R}^n} \chi_{B(0,N)} f(x) \int_{\mathbb{R}^n} \varphi(\xi) e^{-2\pi i x \cdot \xi} d\xi dx \\ &= \int_{\mathbb{R}^n} \chi_{B(0,N)} f(x) \widehat{\varphi}(x) dx \rightarrow \int_{\mathbb{R}^n} f(x) \widehat{\varphi}(x) dx = \langle f, \widehat{\varphi} \rangle, \end{aligned}$$

where the convergence at the end follows from the dominated convergence theorem.

2.3.8. Suppose $|c_k| \leq (1 + |k|)^M$ for $k \in \mathbb{Z}^n$ and some $M > 0$. Let $f \in \mathcal{S}$. We show that

$$\sum_{|k| \leq N} c_k \delta_k$$

converges to a distribution in $\mathcal{S}'$ as $N \rightarrow \infty$. First choose $R > 0$ large enough so that $\sum_{k \in \mathbb{Z}^n} (1 + |k|)^{M-R}$ is summable. Moreover, there exists $C > 0$ such that $|f(x)| \leq C(1 + |x|)^{M-R}$. Then for $f \in \mathcal{S}$, we have

$$\langle \sum_{|k| \leq N} c_k \delta_k, f \rangle = \sum_{|k| \leq N} c_k f(k).$$

We show the last series is in fact absolutely convergent, which implies the original sum converges. We can write

$$\sum_{|k| \leq N} |c_k| |f(k)| \leq \sum_{|k| \leq N} (1 + |k|)^M C (1 + |k|)^{-R} = C \sum_{|k| \leq N} (1 + |k|)^{M-R}$$

which converges as $N \rightarrow \infty$. Then the original sum converges to $\sum_{k \in \mathbb{Z}^n} c_k f(k) = \langle \sum_{k \in \mathbb{Z}^n} c_k \delta_k, f \rangle$.

The Fourier transform of a distribution is defined by $\langle \hat{u}, \varphi \rangle := \langle u, \hat{\varphi} \rangle$. We can write

$$\langle \hat{u}, \varphi \rangle = \langle u, \hat{\varphi} \rangle = \lim_{N \rightarrow \infty} \langle \sum_{|k| \leq N} c_k \delta_k, \hat{\varphi} \rangle = \lim_{N \rightarrow \infty} \langle \sum_{|k| \leq N} c_k e^{-2\pi i \xi \cdot k}, \varphi \rangle,$$

and thus the sequence of functions $\sum_{|k| \leq N} c_k e^{-2\pi i \xi \cdot k}$ converges to $\hat{u}$ in the weak* topology on $\mathcal{S}'$.